Raw Thoughts on AI Pair Programming Tools
Compiled by Jolly Gupta
Yes, Claude helped 😄

What is Pair Programming?

Old way — Two developers sit together, one codes, one reviews in real time.

New way — **You + AI = Pair Programming** AI sits beside you, suggests code, fixes errors, reviews, explains — in real time.

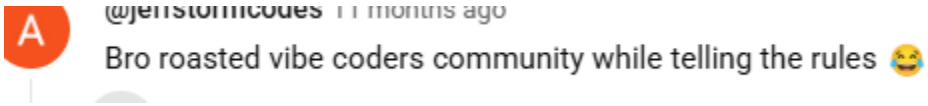
TOP PAIR PROGRAMMING AI TOOLS:

TIER 1 — BEST IN CLASS:

1. Cursor AI

- Best overall AI code editor
- Understands your ENTIRE codebase at once
- You describe feature — it writes across multiple files
- Chat with your code — ask questions about it
- Auto fixes errors automatically
- Composer mode — builds entire features from description
- **Best for — Full project pair programming**

Vibe Coding https://www.youtube.com/watch?v=oFAPQv5UikM



2. GitHub Copilot

- Lives inside VS Code, JupyterLab, PyCharm
- Suggests next line as you type
- Tab to accept — feels like telepathy
- Copilot Chat — ask questions about your code
- Works with all languages
- **Best for — Line by line coding assistance**

3. Claude

- Paste your entire code — it understands fully
- Best at debugging complex ML errors
- Cleanest most production ready code suggestions
- Explains WHY not just WHAT
- Best for long complex codebases
- **Best for — Deep code analysis and debugging**

TIER 2 — VERY GOOD:

4. ChatGPT

- Strong coding across all languages
- Great for quick code generation
- Good at explaining code simply
- GPT4o very fast responses
- **Best for — Quick snippets and explanations**

5. Amazon Code Whisperer

- Free for individual developers
- Works inside VS Code and JupyterLab
- Strong AWS integration
- Good Python and SQL support
- **Best for — AWS cloud and data engineering**

6. Tabnine

- AI autocomplete inside your IDE
- Privacy focused — can run locally
- Learns YOUR coding style over time
- **Best for — Privacy conscious developers**

7. Codeium

- Free alternative to GitHub Copilot
- Works in 40+ IDEs

- Fast autocomplete suggestions
- **Best for — Budget conscious developers**

TIER 3 — SPECIALIZED:

8. Replit AI

- Built into Replit environment
- Great for quick prototyping
- Explains and fixes code inline
- **Best for — Quick experiments**

9. Sourcegraph Cody

- Understands large enterprise codebases
- Searches across entire repositories
- **Best for — Large team codebases**

10. Aider

- Open source AI pair programmer
- Works in terminal
- Connects to Claude or GPT
- Makes changes directly to your files
- **Best for — Terminal lovers and open source fans**

SPECIFICALLY FOR DATA SCIENTISTS / ML ENGINEERS:

For Jupyter Notebooks:

Tool	How it Helps
GitHub Copilot	Works directly inside JupyterLab
Claude	Paste notebook cells, get full analysis
ChatGPT	Quick cell by cell assistance
Amazon CodeWhisperer	Free Jupyter integration
Jupyter AI	Built in AI directly inside Jupyter

For ML Code Specifically:

Task	Best Tool
Writing PyTorch training loops	Claude or DeepSeek
Debugging model not converging	Claude
Writing Pandas data pipelines	Claude or Copilot
SQL queries for data	ChatGPT or Claude
Sklearn pipelines	Copilot or Claude
Writing FastAPI for model serving	Claude
Docker for ML deployment	ChatGPT or Claude
Feature engineering	Claude or ChatGPT
Data visualization code	ChatGPT or Copilot
Writing unit tests for ML code	Claude

For VS Code Users:

Best Setup —

VS Code

+ GitHub Copilot (inline suggestions)

+ Claude via API (deep analysis)

+ Cursor AI (full project understanding)

For PyCharm Users:

Best Setup —

PyCharm

+ GitHub Copilot plugin

+ ChatGPT for quick help

+ Claude for complex debugging

For Terminal / Command Line:

Best Setup —

Aider + Claude

Run in terminal

AI makes direct file changes

Perfect for ML pipeline work

COMPARISON TABLE:

Tool	Where it Works	Best For	Free?
Cursor AI	Own editor	Full project	Free limited
GitHub Copilot	VS Code, Jupyter	Line suggestions	Paid $10/mo
Claude	Browser + API	Deep analysis	Free limited
ChatGPT	Browser + API	Quick help	Free limited
CodeWhisperer	VS Code, Jupyter	AWS projects	Free
Codeium	40+ IDEs	Autocomplete	Free
Tabnine	Most IDEs	Private coding	Free limited
Aider	Terminal	Open source	Free
Replit AI	Replit	Quick prototypes	Free limited
Continue.dev	VS code	100% Private local	✅ Free

THE PERFECT PAIR PROGRAMMING SETUP:

For DS/ML Engineers — Ultimate Stack:

Cursor AI

(Main editor — understands full codebase)

+

GitHub Copilot

(Real time line suggestions while typing)

+

Claude

(Deep debugging and complex analysis)

+

Groq

(Ultra fast quick questions)

+

Perplexity

(Find latest library documentation)

Real Workflow Example:

Writing a ML Pipeline —

1. Open Cursor AI
   → Describe the pipeline in plain English
   → It generates the full structure

2. GitHub Copilot
   → Autocompletes as you refine details
   → Tab to accept suggestions

3. Hit an error?
   → Paste into Claude
   → Claude explains and fixes deeply

4. Need quick syntax?

→ Ask Groq

→ Answer in milliseconds

5. Need latest library docs?

→ Search Perplexity

→ Current documentation instantly

Multi.ai for Pair Programming:

Remember Multi.ai we discussed earlier? It enables HUMAN + HUMAN pair programming remotely —

Multi.ai = Two humans, one screen, both control (Privacy concern So learner can use)

AI Tools = One human, AI as second programmer

Best combo — You + colleague on Multi.ai + Cursor AI + Claude = Three way pair programming — Human + Human + AI

For Privacy Sensitive ML Code:

Tool	Privacy Level
Tabnine Local	Runs on your machine
Aider + Ollama	Fully local, zero cloud
Cursor with local model	Private mode available
Continue.dev + Ollama	Open source, fully local

Continue.dev — special mention —

- Open source GitHub Copilot alternative
- Connects to local Ollama models
- Zero data sent anywhere
- Free forever
- Perfect for sensitive enterprise ML code

Bottom line:

Need	Use
Best overall pair programmer	Cursor AI + Claude
Best inline suggestions	GitHub Copilot
Best for Jupyter notebooks	Copilot + Claude
Best free option	Codeium + Claude free
Best for privacy	Continue.dev + Ollama
Best for debugging	Claude
Best for speed	Groq
Best for full project	Cursor AI

One line summary:

GitHub Copilot types with you. Cursor AI thinks with you. Claude debugs with you. Together they make you a 10x Data Scientist and ML Engineer.